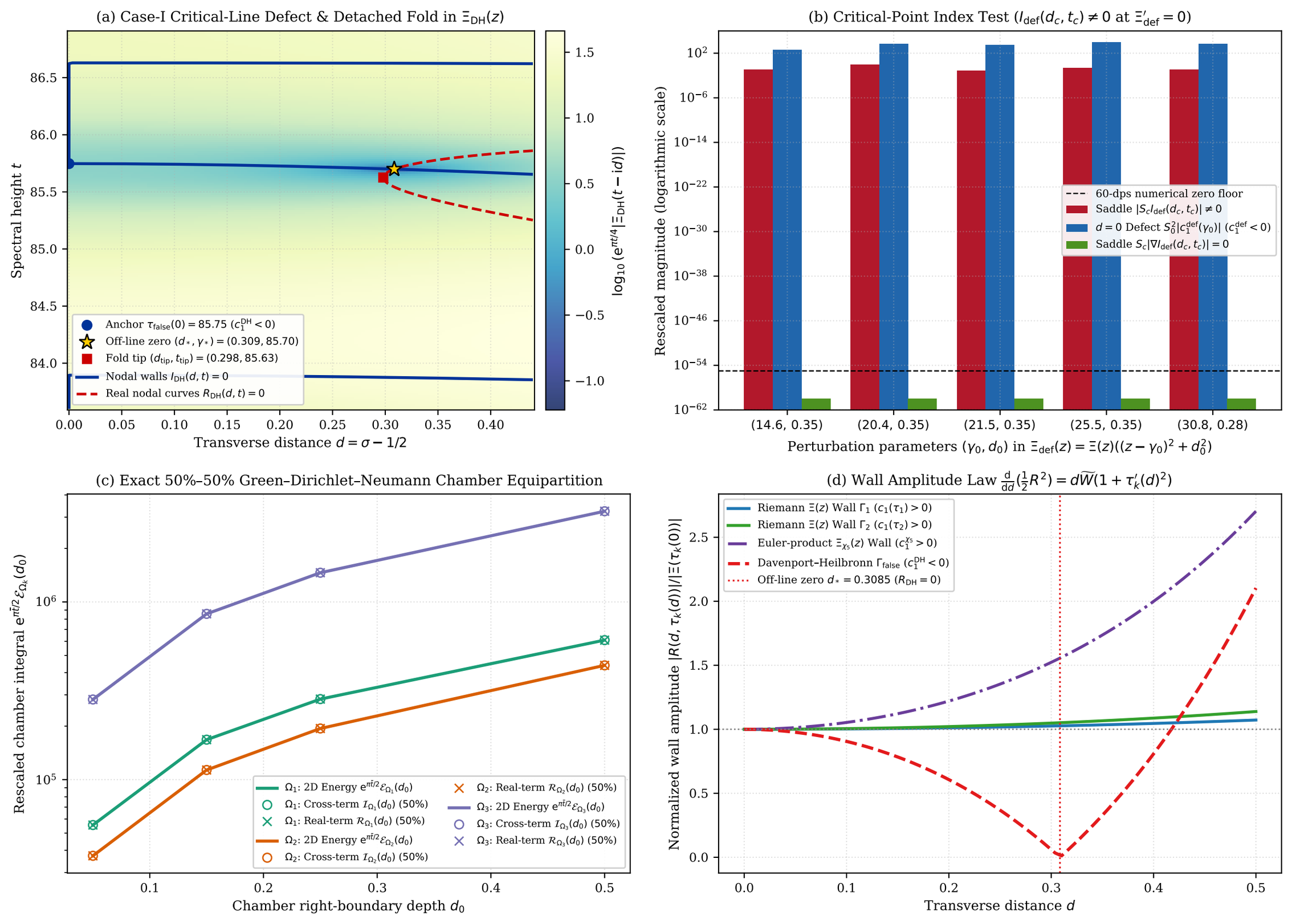

(a) Case-I Critical-Line Defect & Detached Fold in $\Xi_{\mathrm{DH}}(z)$
Spectral height t
Transverse distance $d = \sigma - 1/2$
$\log_{10}(e^{\pi t/4}|\Xi_{\mathrm{DH}}(t - id)|)$
Anchor $\tau_{\mathrm{false}}(0) = 85.75$ ($c_1^{\mathrm{DH}} < 0$)
Off-line zero $(d_*, \gamma_*) = (0.309, 85.70)$
Fold tip $(d_{\mathrm{tip}}, t_{\mathrm{tip}}) = (0.298, 85.63)$
Nodal walls $I_{\mathrm{DH}}(d, t) = 0$
Real nodal curves $R_{\mathrm{DH}}(d, t) = 0$
(b) Critical-Point Index Test ($I_{\mathrm{def}}(d_c, t_c) \neq 0$ at $\Xi'_{\mathrm{def}} = 0$)
Rescaled magnitude (logarithmic scale)
60-dps numerical zero floor
Saddle $|S_c I_{\mathrm{def}}(d_c, t_c)| \neq 0$
$d = 0$ Defect $S_0^2 |c_1^{\mathrm{def}}(\gamma_0)|$ ($c_1^{\mathrm{def}} < 0$)
Saddle $S_c |\nabla I_{\mathrm{def}}(d_c, t_c)| = 0$
(14.6, 0.35)
(20.4, 0.35)
(21.5, 0.35)
(25.5, 0.35)
(30.8, 0.28)
Perturbation parameters (γ_0, d_0) in $\Xi_{\mathrm{def}}(z) = \Xi(z)((z - \gamma_0)^2 + d_0^2)$
(c) Exact 50%–50% Green–Dirichlet–Neumann Chamber Equipartition
Rescaled chamber integral $e^{\pi t/2}\mathcal{E}_{\Omega_k}(d_0)$
Ω_1: 2D Energy $e^{\pi t/2}\mathcal{E}_{\Omega_1}(d_0)$
Ω_1: Cross-term $\mathcal{I}_{\Omega_1}(d_0)$ (50%)
Ω_1: Real-term $\mathcal{R}_{\Omega_1}(d_0)$ (50%)
Ω_2: 2D Energy $e^{\pi t/2}\mathcal{E}_{\Omega_2}(d_0)$
Ω_2: Cross-term $\mathcal{I}_{\Omega_2}(d_0)$ (50%)
Ω_2: Real-term $\mathcal{R}_{\Omega_2}(d_0)$ (50%)
Ω_3: 2D Energy $e^{\pi t/2}\mathcal{E}_{\Omega_3}(d_0)$
Ω_3: Cross-term $\mathcal{I}_{\Omega_3}(d_0)$ (50%)
Ω_3: Real-term $\mathcal{R}_{\Omega_3}(d_0)$ (50%)
Chamber right-boundary depth d_0
(d) Wall Amplitude Law $\frac{\mathrm{d}}{\mathrm{d}d}(\frac{1}{2}R^2) = d\widetilde{W}(1 + \tau_k'(d)^2)$
Riemann $\Xi(z)$ Wall Γ_1 ($c_1(\tau_1) > 0$)
Riemann $\Xi(z)$ Wall Γ_2 ($c_1(\tau_2) > 0$)
Euler-product $\Xi_{\chi_5}(z)$ Wall ($c_1^{\chi_5} > 0$)
Davenport–Heilbronn Γ_{false} ($c_1^{\mathrm{DH}} < 0$)
Off-line zero $d_* = 0.3085$ ($R_{\mathrm{DH}} = 0$)
Normalized wall amplitude $|R(d, \tau_k(d))|/|\Xi(\tau_k(0))|$
Transverse distance d